CYBR 550 – Assignment 1

Enterprise Asset Management Solution Proposal: Axonius

Prepared by: Maxine Jones

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Executive Pitch

Axonius provides Northwestern Medicine with a unified, real-time view of every asset across the enterprise, addressing one of the most persistent and costly gaps in cybersecurity: the lack of complete visibility. Northwestern Medicine's current environment relies on multiple disconnected tools, each offering only partial insight into devices, users, cloud workloads, and SaaS applications. This fragmentation creates blind spots that adversaries exploit and auditors consistently identify as deficiencies. Axonius consolidates data from existing systems into a single authoritative inventory, enabling leadership to reduce cyber risk, strengthen compliance, and eliminate redundant technology. The platform delivers immediate operational value and long-term strategic benefits, making it a necessary investment for a mature, defensible cybersecurity program.

Tool Overview

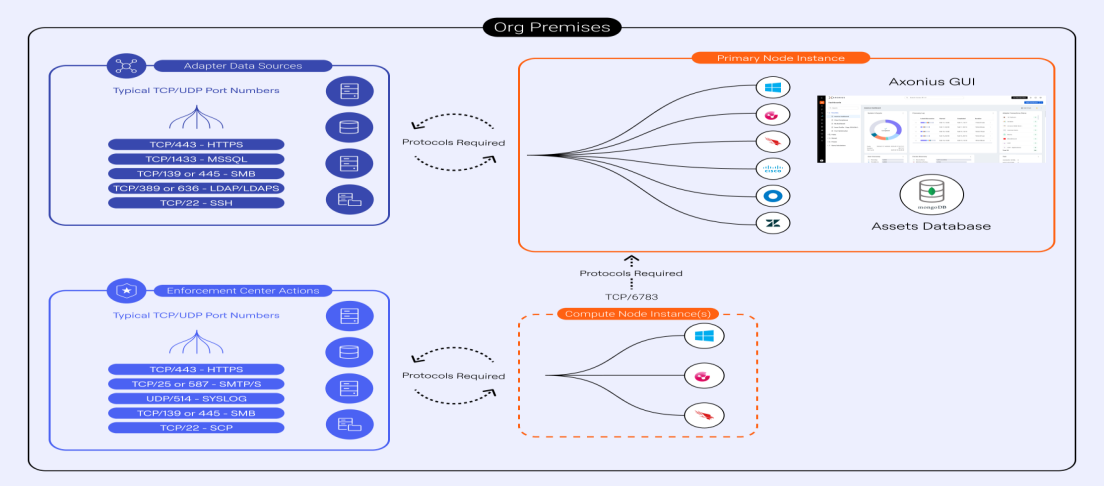

Figure 1: Axonius architecture showing how adapter data sources integrate with the primary node.

As shown in Figure 1, Axonius aggregates data from multiple sources into a unified asset inventory. Axonius is an enterprise-grade cybersecurity asset management and SaaS management platform designed to provide organizations with a complete, accurate, and

continuously updated inventory of all assets. Traditional asset management tools often depend on a single data source, require agents, or provide limited visibility. Axonius takes a different approach by aggregating and correlating data from more than one thousand integrations, enabling comprehensive visibility across hardware, software, cloud environments, SaaS applications, and user identities.

The platform ingests metadata from endpoint security tools, cloud providers, identity systems, MDM/UEM platforms, vulnerability scanners, and network infrastructure. It then correlates this information to create a single, deduplicated asset record for each device, user, or application. This unified inventory becomes the foundation for automated security gap detection, compliance validation, and operational workflows. Axonius does not require agents or intrusive scanning, allowing organizations to leverage the tools they already own without additional overhead.

Axonius manages a broad range of asset categories. Hardware assets include laptops, desktops, servers, mobile devices, virtual machines, network equipment, and IoT or OT systems. Software assets include installed applications, security agents, and configuration data. Cloud assets span AWS, Azure, and GCP workloads, IAM identities, and containerized environments. SaaS assets include both sanctioned and unsanctioned applications, enabling organizations to identify shadow IT and evaluate license utilization. Identity assets include user accounts, privileged access, authentication sources, and role mappings. This breadth allows Axonius to serve as a single source of truth for the entire technology environment.

Data collection is performed through Axonius connectors, which integrate with existing systems through APIs. Examples include CrowdStrike and SentinelOne for endpoint security, Intune and Jamf for device management, Okta and Azure AD for identity, AWS and Azure for cloud workloads, Qualys and Tenable for vulnerability scanning, Cisco ISE for network access control, and ServiceNow for IT service management. Axonius correlates this data using deterministic and probabilistic matching techniques to produce validated asset records. This approach eliminates duplicate entries, resolves conflicting information, and ensures that asset data remains accurate and current.

Several features distinguish Axonius from other asset management platforms. Its unified asset inventory consolidates data from all tools into a single authoritative source, eliminating blind spots and inconsistent records. Automated security gap detection identifies missing controls, misconfigurations, unmanaged devices, and policy violations. Automated enforcement workflows allow Axonius to trigger actions such as disabling accounts, opening tickets, or sending alerts, reducing manual work and improving response time. SaaS management capabilities identify all applications in use, including unsanctioned ones, helping organizations reduce cost, improve compliance, and prevent data leakage. The platform's rapid deployment model allows most organizations to achieve full functionality in less than a day, as it relies on existing tools rather than new infrastructure.

Axonius was selected because it is industry-leading, widely adopted, and purpose-built for cybersecurity. It is vendor-agnostic, highly scalable, and aligned with zero-trust principles and compliance frameworks such as NIST, ISO 27001, HIPAA, and PCI. The platform provides immediate value by improving visibility and reducing operational burden, while also supporting long-term strategic goals related to risk management and governance. Its ability to unify data from disparate systems makes it one of the most complete and realistic solutions for enterprise asset management available today.

Business Framing

Asset management is foundational to cybersecurity because organizations cannot protect assets they do not know exist. For a large healthcare organization like Northwestern Medicine — managing thousands of clinical endpoints, cloud workloads, and third-party vendor integrations across multiple hospital campuses — this challenge is especially acute. A breach involving unmanaged medical devices or orphaned staff accounts carries

not just financial consequences, but HIPAA liability and patient safety risk

Figure 2: Security blind spots caused by shadow IT, unmanaged devices, and limited visibility across enterprise networks.

As shown in Figure 2, a lack of centralized asset visibility creates significant security blind spots across the enterprise. Unpatched devices, unmanaged cloud workloads, orphaned user accounts, and shadow SaaS applications represent common attack vectors that lead directly to breaches and compliance failures. These issues also create operational inefficiencies, as teams spend significant time reconciling inconsistent data across multiple systems.

Axonius addresses these challenges by providing a single, unified view of every asset across Northwestern Medicine's environment.. For executive leadership, this visibility supports informed decision-making, reduces risk exposure, and improves audit readiness. For security teams, Axonius enables faster incident response, automated compliance validation, and consistent enforcement of security controls. For IT operations, the platform reduces manual work, eliminates redundant tools, and improves coordination across teams.

From a business perspective, Axonius delivers measurable value. It reduces risk by identifying unmanaged devices and missing security controls. It improves operational efficiency by automating asset discovery, reporting, and remediation workflows. It optimizes cost by identifying unused SaaS licenses and redundant technology. It strengthens compliance by providing accurate, evidence-based reporting aligned with regulatory frameworks. Axonius transforms asset management from a manual, error-prone

process into a strategic capability that supports organizational resilience and long-term cybersecurity maturity.

References

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